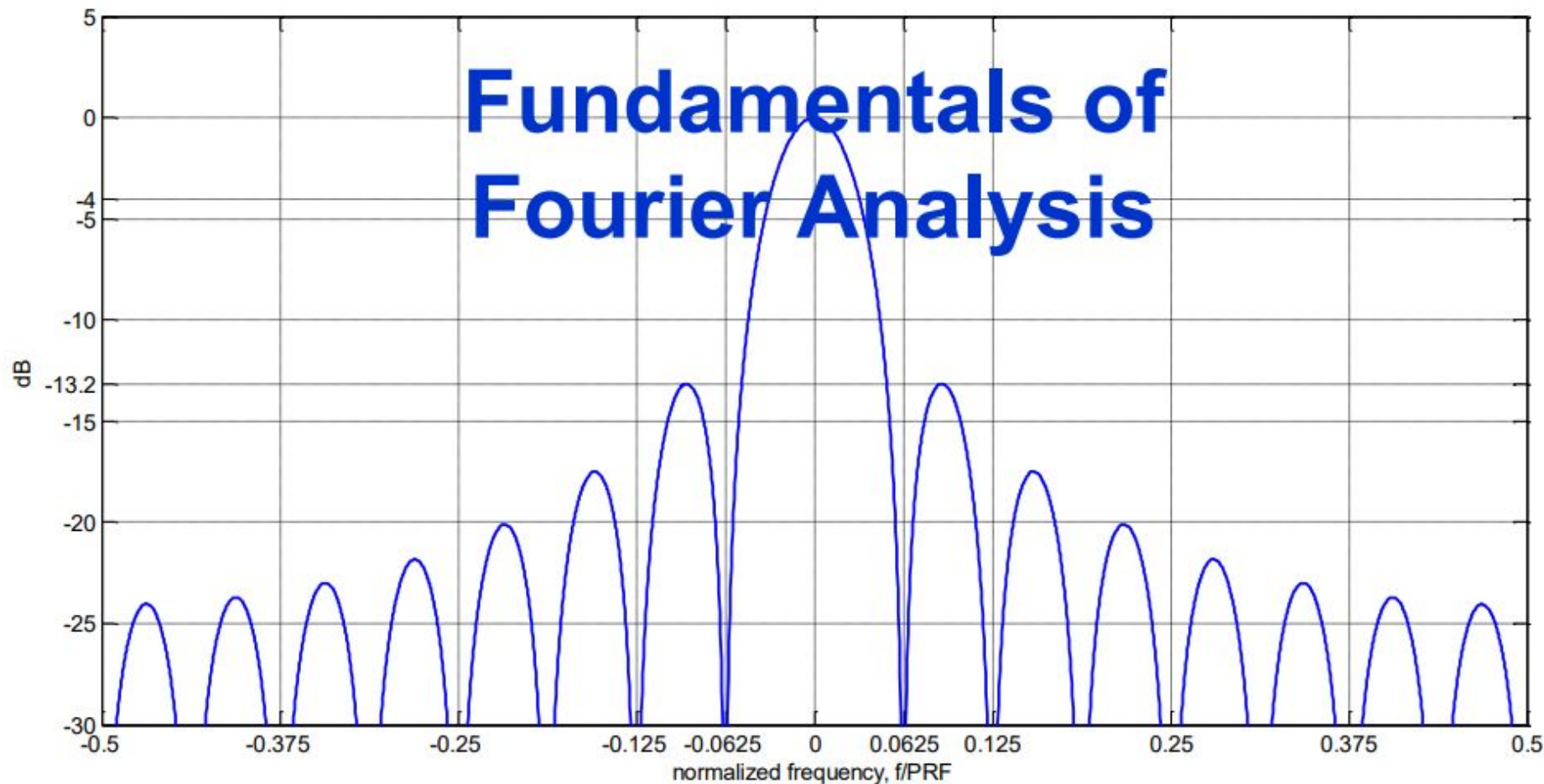


Fundamentals of Fourier Analysis

EE417 / EE517
CUA Spring 2024

Tuesdays 7:00-9:30pm
Instructor: Prof. Kevin Russo

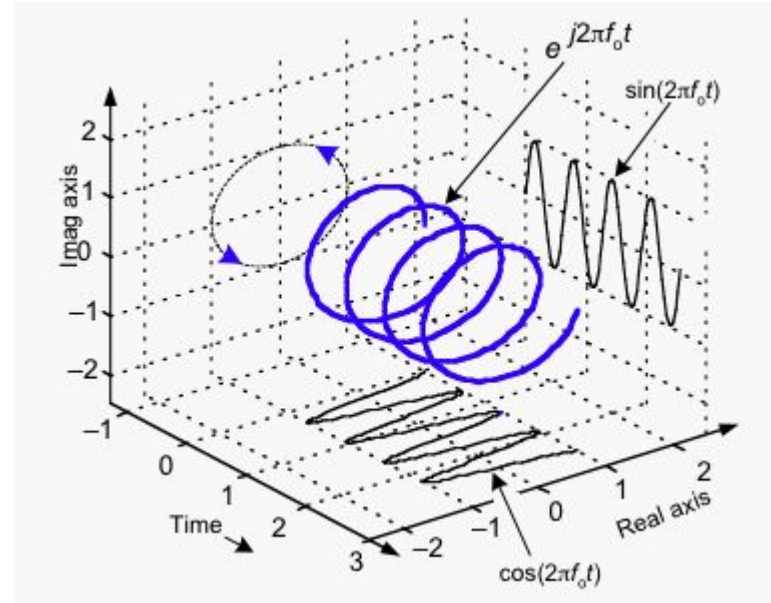
Fundamentals of Fourier Analysis



The Fourier Transform

But first, the Fourier Series

Represents arbitrary functions as infinite sums of sines and cosines



The (Continuous) Fourier Transform

- The definition we will use is:

- “Forward transform”:

$$X(\Omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\Omega t} dt$$

- “Inverse transform”:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\Omega) e^{+j\Omega t} d\Omega$$

- Typically t in seconds, Ω in radians/sec

- If we don't qualify the name, “Fourier transform” means the *continuous* F.T.

The Continuous Fourier Transform in Cyclical Frequency Units

- **Alternatively, using $\Omega = 2\pi F$, F in hertz (Hz)**

- **Forward transform:**

$$X(F) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi Ft} dt$$

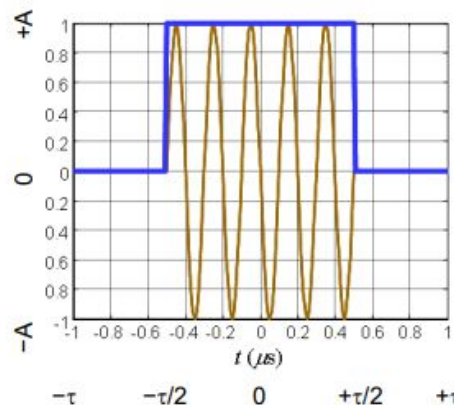
- **Inverse transform:**

$$x(t) = \int_{-\infty}^{+\infty} X(F) e^{+j2\pi Ft} dF$$

- Notice that there is no $1/2\pi$ scale factor on the inverse transform in this version

The Fourier Transform of a Complex Sinusoid of **Finite** Duration

- **Let**
$$x(t) = \begin{cases} A \exp(j\Omega_0 t), & |t| \leq \tau/2 \\ 0, & \text{otherwise} \end{cases}$$



Sinc and Box
are FT pairs

- Then “it can be shown” that

$$X(F) = A\tau \frac{\sin[\pi\tau(F - F_0)]}{\pi\tau(F - F_0)} \equiv A\tau \operatorname{sinc}[\pi\tau(F - F_0)]$$

$$\left(\operatorname{sinc}(z) \equiv \frac{\sin(\pi z)}{\pi z} \right)$$

$$\operatorname{sinc} 0 := \lim_{x \rightarrow 0} \frac{\sin(ax)}{ax} = 1$$

$$\frac{\sin \pi x}{\pi x} = 1 - \frac{\pi^2 x^2}{3!} + \frac{\pi^4 x^4}{5!} - \frac{\pi^6 x^6}{7!} + \dots$$

Fourier Transform work-out

$$X(F) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi Ft} dt$$

$$x(t) = \begin{cases} A \exp(j\Omega_0 t), & |t| \leq \tau/2 \\ 0, & \text{otherwise} \end{cases}$$

$$X(F) = A\tau \operatorname{sinc}[\pi\tau(F - F_0)]$$

Fourier Transform work-out

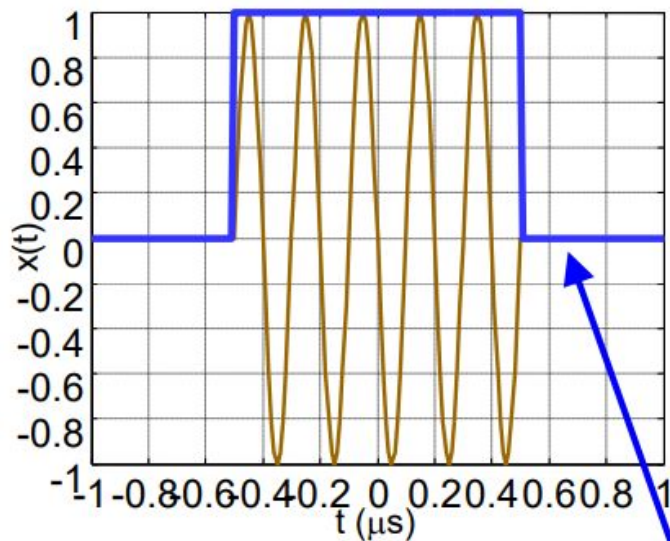
Calculate where these features occur:

Peak Value _____

Zero-Crossings _____

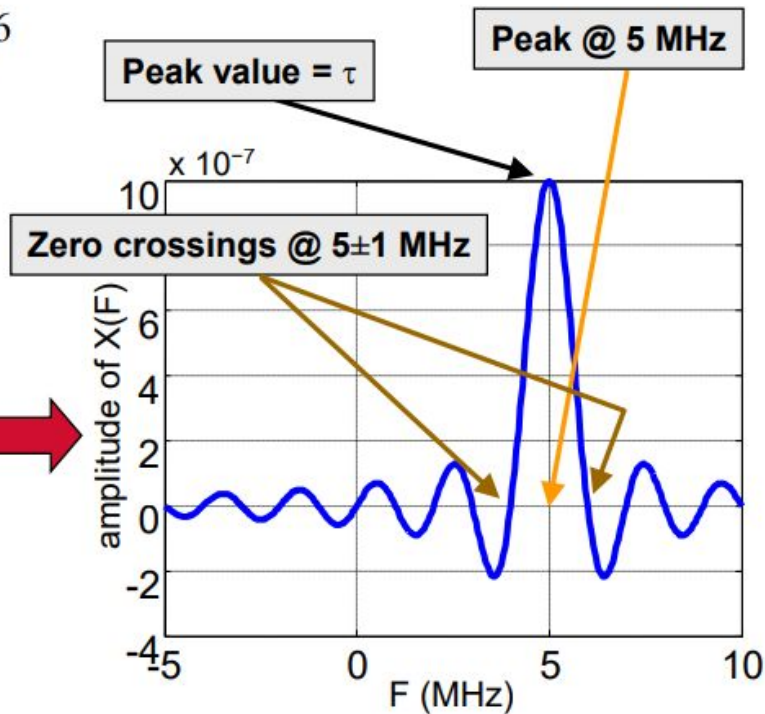
Example: A 5 MHz Finite-Length Pulse

- $\tau = 1 \mu\text{s}$, $\Omega_0 = 10\pi \times 10^6$
($F_0 = 5 \text{ MHz}$)



$x(t)$

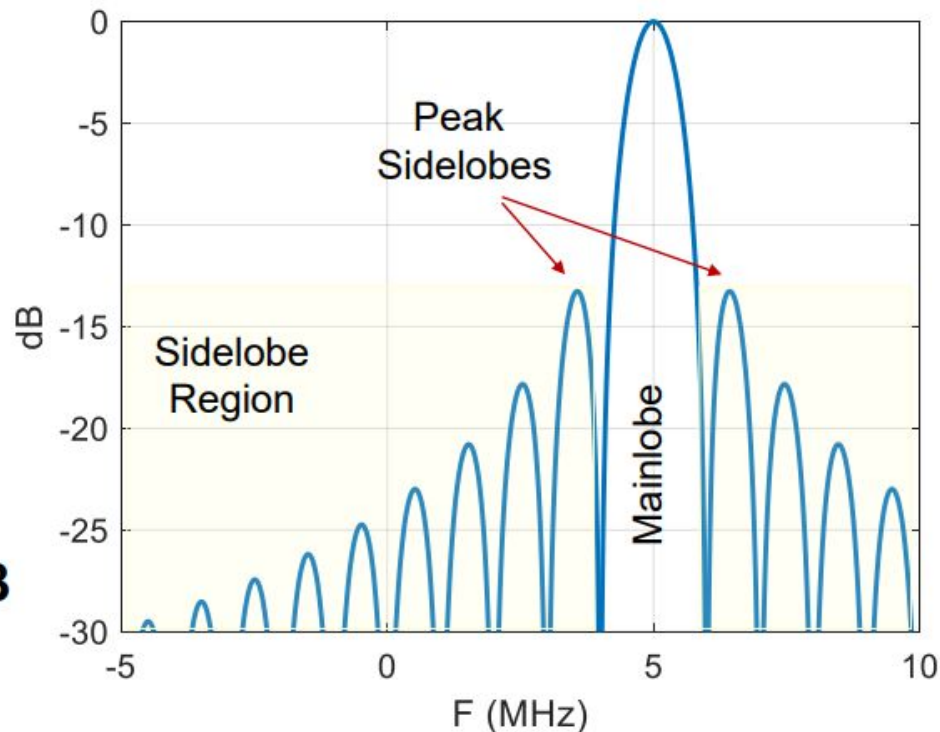
Envelope of
complex sinusoid



$X(F)$

Properties of the Sinc Function: Sidelobes

- **Peak occurs at**
 Ω_0
 $(F_0 = \Omega_0/2\pi)$
- **Sidelobes**
 - **Peak: -13.26 dB**
 - **Rolloff: $\sim 1/F$**



Properties of the Sinc Function: Mainlobe Width

- The other important metric is the width of the mainlobe
- Several measures of width are in common use:

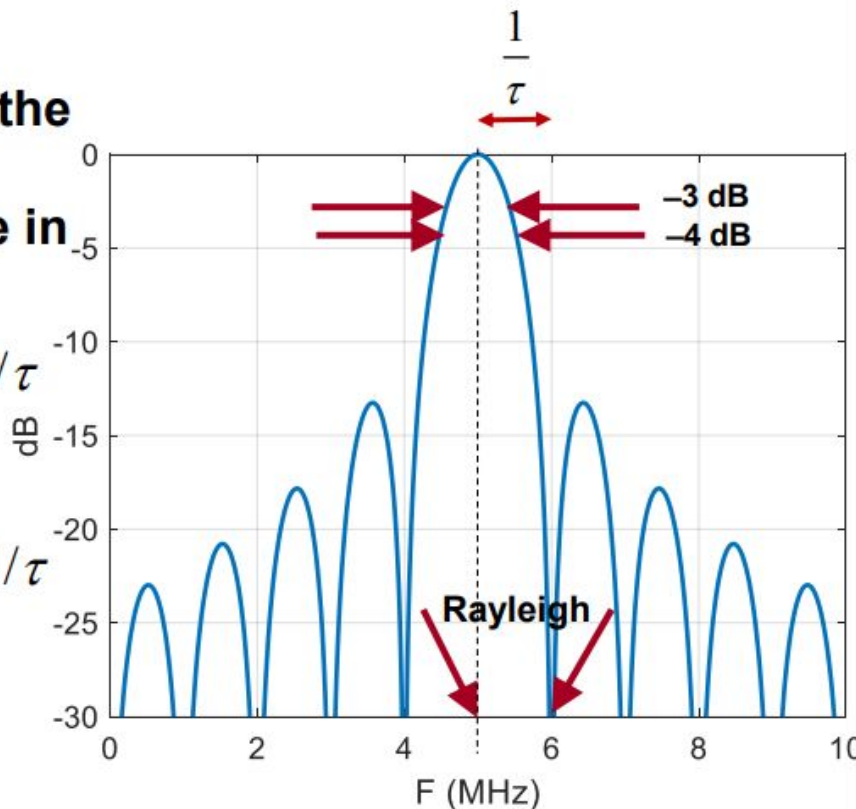
- 3 dB (0.707 of peak): $0.89/\tau$

- 4 dB (0.63 of peak): $1/\tau$

- Peak-to-null (**Rayleigh**): $1/\tau$

- Null-to-null: $2/\tau$

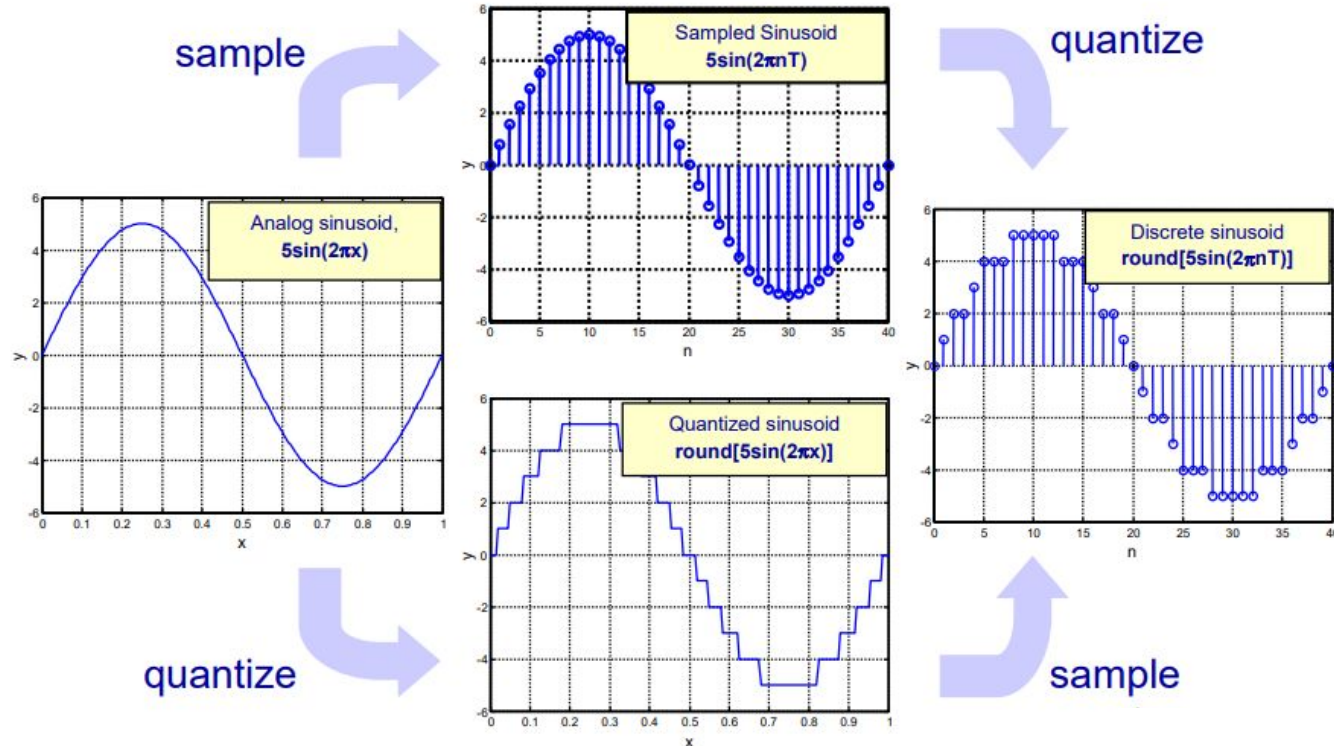
- **All vary as $1/\tau$!**



Sampling

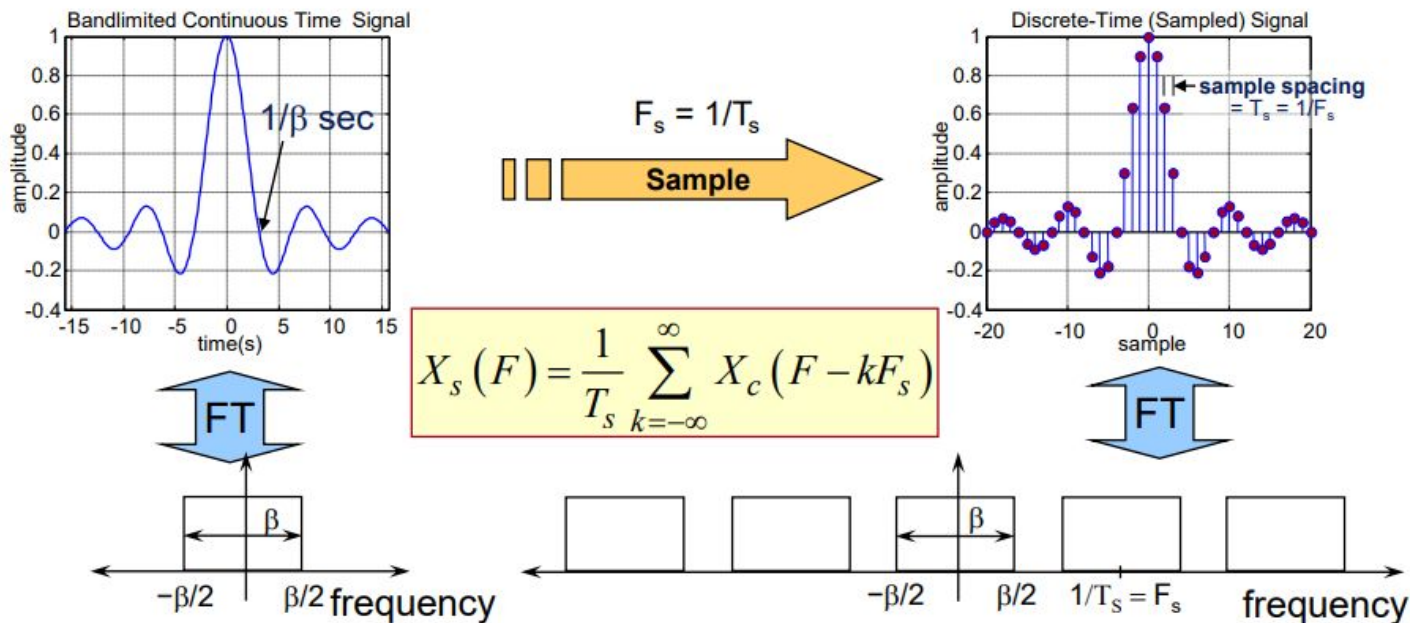
Digital Signals

- A digital signal is quantized in two ways:
 - **Sampled** in the independent variable (time, space, frequency)
 - **Quantized** in amplitude



A Fact of Continuous-Time Fourier Transforms

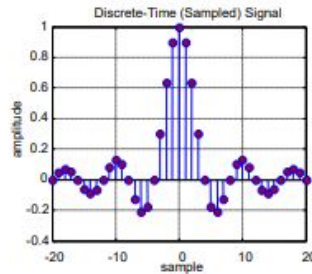
- Sampling a function at an interval of Δt in the time domain causes it to be replicated at intervals of $1/\Delta t$ in the frequency domain:



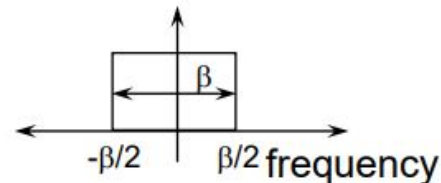
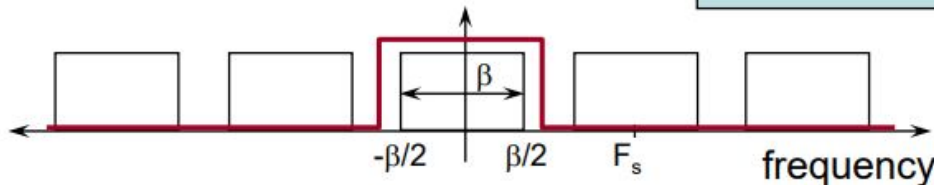
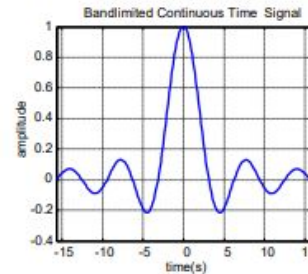
- True whether the function is bandlimited or not
- True whether the function is real or complex
- True whether the function is baseband (spectrum centered at origin) or not

Reconstructing the Original Signal

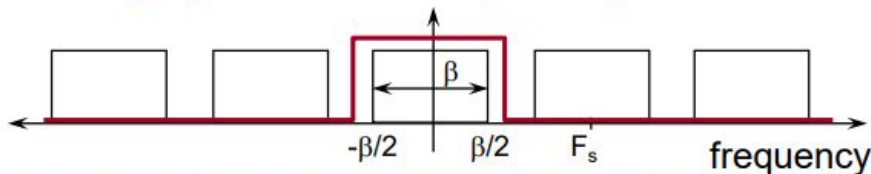
- A signal and its Fourier transform are a unique pair
- If we can reconstruct the original spectrum, we have reconstructed the original signal
- We can reconstruct from a sampled signal (replicated spectrum) using a low pass filter:



Lowpass Filter



The Nyquist Sampling Theorem



- This reconstruction process will work if the Fourier spectrum replications do not overlap, which establishes **two conditions**:
 - The signal must be bandlimited
 - The replicas must not overlap →

$$\beta / 2 < F_s - \beta / 2 \rightarrow F_s = \frac{1}{T_s} > 2(\beta / 2) = \beta$$

(Nyquist Sampling Rate)

- So the sampling rate must be at least the **full bandwidth of the signal spectrum**
- This is equivalent to “**twice the highest frequency**” if the signal is at baseband
 - The “full bandwidth” rule is more general
- A complex signal requires β **complex** samples/second, or 2β real samples/second

The Discrete Time Fourier Transform (DTFT)

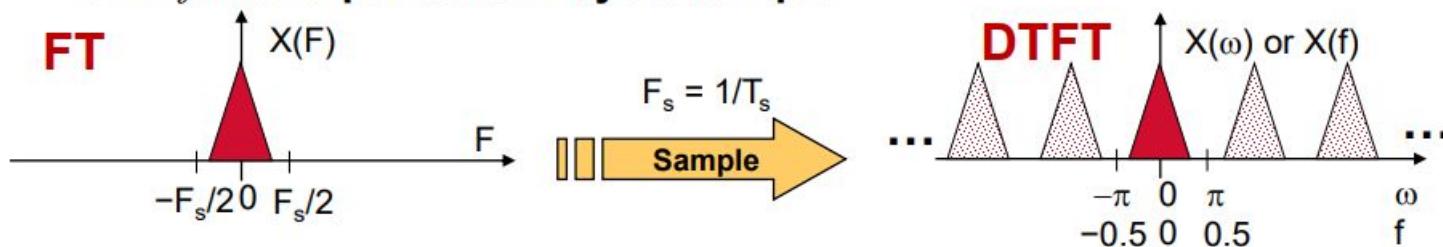
The Discrete-Time Fourier Transform

- The DTFT is the Fourier transform for a *discrete-time* (i.e. “digital”) signal $x[n]$:

$$X(\omega) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$$

$$X(f) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j2\pi f n}$$

- The normalized frequency variable ω (radians/sample) or f (cycles/sample) is **continuous**
- The spectrum is **periodic** in ω with a period of 2π radians/sample, or in f with a period of 1 cycle/sample



- The inverse DTFT is

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{+\pi} X(\omega) e^{+j\omega n} d\omega$$

$$x[n] = \int_{-0.5}^{+0.5} X(f) e^{+j2\pi f n} df$$

Normalized Frequency

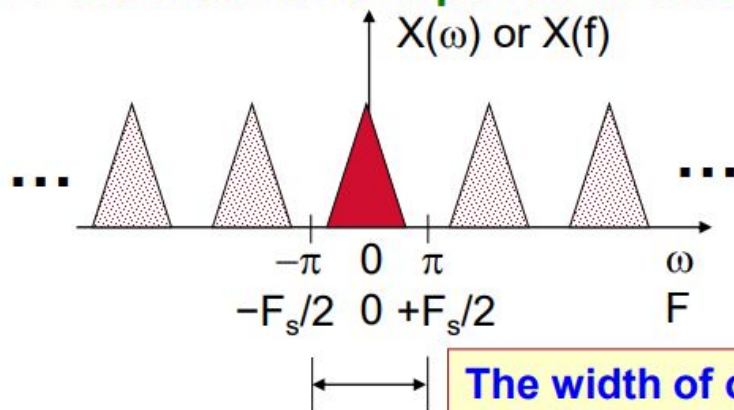
- f and ω are “**normalized**” or “**digital**” frequencies
 - In units of cycles/sample or radians/samples
 - Just called “cycles” or “samples” in some texts
- To convert normalized frequencies back to engineering units (hertz (Hz) or radians/second)

$$\omega = \Omega T = \frac{\Omega}{F_s}, \quad \Omega = \omega F_s$$

$$f = FT = \frac{F}{F_s}, \quad F = fF_s$$

Frequency Span of the DTFT

- The principal period of a DTFT covers a range in ω of $(-\pi, +\pi)$
- Earlier we saw that analog and normalized (“digital”) frequencies are related by $F = fF_s = \omega F_s / 2\pi$
- This means that when you do the DTFT of discrete data that was collected at a sampling rate of F_s samples/sec,
- **The principal period of the spectrum always covers the range $(-F_s/2, +F_s/2)$ Hz**
- **And the width of one period of the spectrum is always F_s Hz!**



The width of one period is F_s Hertz!

The Discrete-Time Fourier Transform of a Complex Sinusoid

- The slow-time receiver output of an ideal, constant-radial-velocity moving target for an N -pulse coherent processing interval (CPI) is a complex sinusoid:

$$x[n] = Ae^{j2\pi F_d nT} = Ae^{j2\pi f_d n}, \quad n = 0, \dots, N-1, \quad f_d = F_d T$$

- The DTFT that results is a “digital sinc” or “aliased sinc” (“asinc”) or “Dirichlet” function:

$$X(f) = A \frac{\sin[\pi(f - f_d)N]}{\sin[\pi(f - f_d)]} e^{-j\pi(N-1)(f - f_d)}$$

- Very similar to the analog case, but not identical

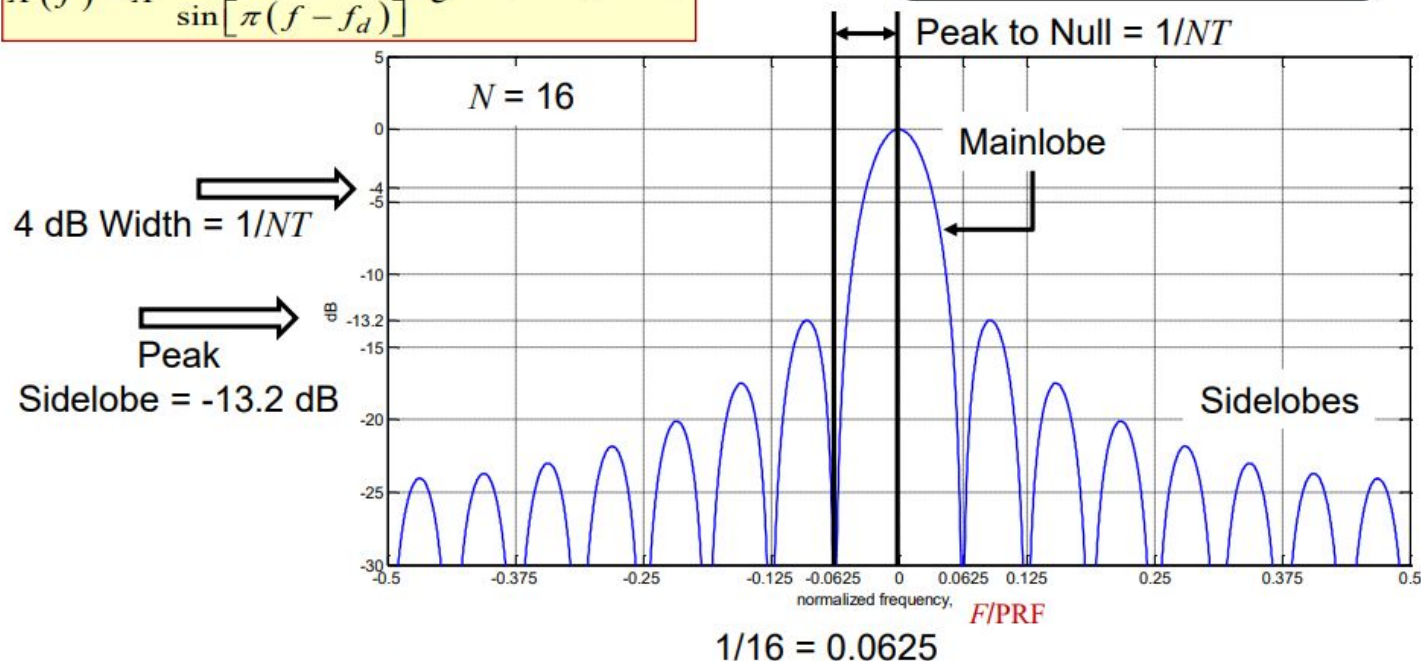
The Dirichlet Response (dB Scale)

- Consists of mainlobe and sidelobe response
- Peak sidelobe = -13.2 dB
- Rayleigh frequency resolution = $1/NT = PRF/N$
- 3 dB Width = $0.89/NT$
- 4 dB width = $1/NT$

$$\delta f = \frac{1}{NT} = \frac{PRF}{N}$$

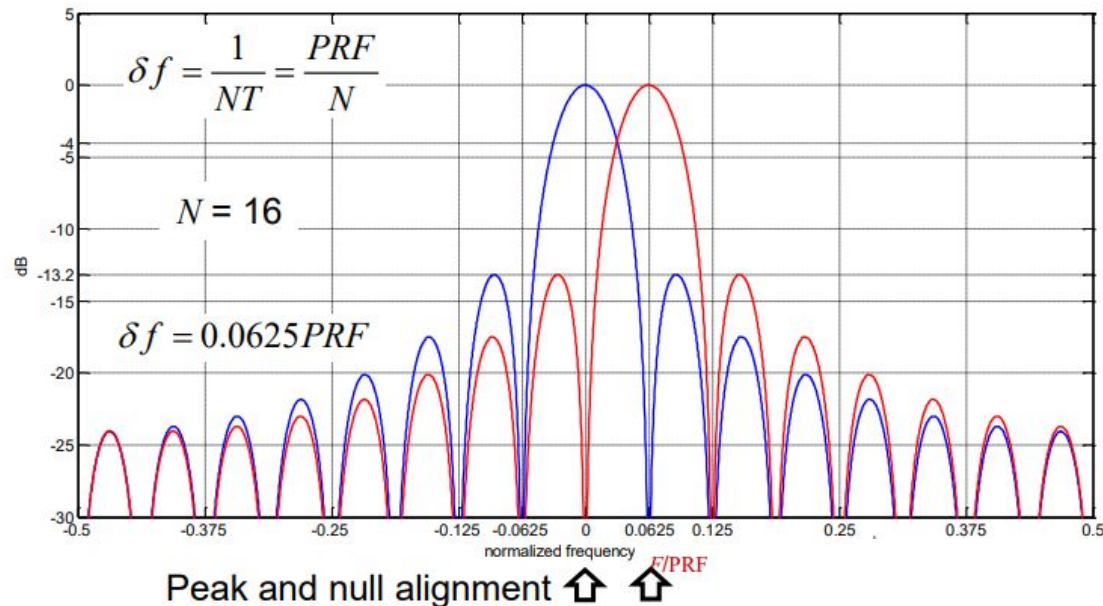
Frequency resolution inversely proportional to dwell time

$$X(f) = A \frac{\sin[\pi(f-f_d)N]}{\sin[\pi(f-f_d)]} e^{-j\pi(N-1)(f-f_d)}$$



Rayleigh Resolved Doppler Shifts

- Based on the Rayleigh criterion, two Doppler shifts are resolved if the peak of one aligns with the first null of the second
- This is equivalent to the -4 dB width separation
- What determines the shape of the composite response?



Constructive Addition

- Each Doppler return has an associated range dependent, constant phase

$$\phi_0 = 2\pi \frac{2R_0}{\lambda}$$

- If the phase difference between the two targets is an **integer multiple of 2π** , the two responses **add constructively**

Target #1
(Blue)

$$\phi_1 = 2\pi \frac{2R_1}{\lambda}$$

Target #2
(Red)

$$\phi_2 = 2\pi \frac{2R_2}{\lambda}$$

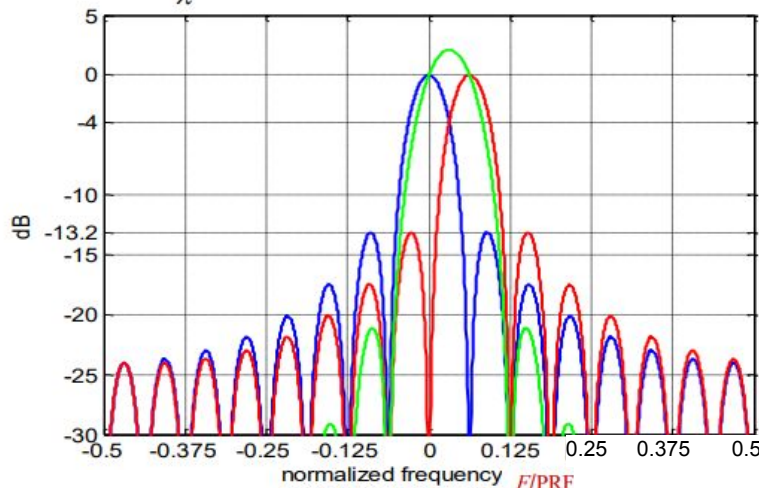
- In this instance, the response doesn't exhibit a null or dip between the two returns
- Visually difficult to determine if two Doppler shifts are present

Phase Difference

$$\Delta\phi = 2\pi \frac{2(R_1 - R_2)}{\lambda}$$

Same Range Bin

$$|R_1 - R_2| \leq \delta r$$



Destructive Addition

- If the phase difference between the two targets is an **odd multiple of π** , the two responses **add destructively**

Target #1
(Blue)
 $\phi_1 = 2\pi \frac{2R_1}{\lambda}$

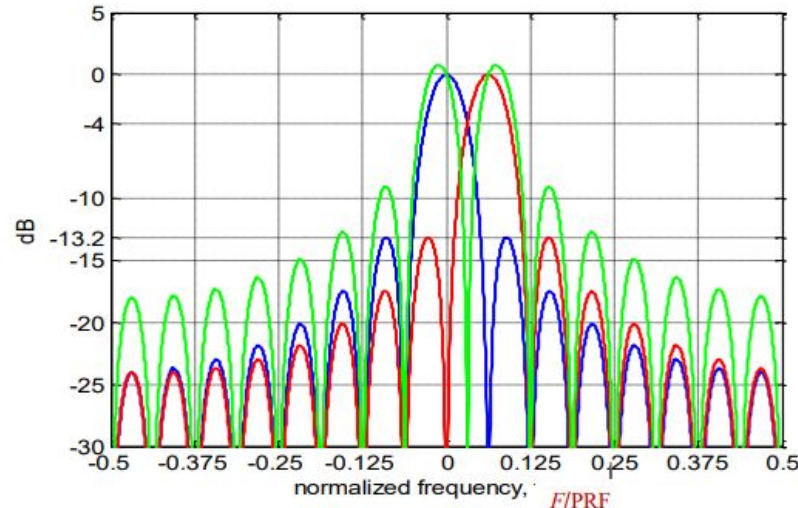
Target #2
(Red)
 $\phi_2 = 2\pi \frac{2R_2}{\lambda}$

Phase Difference
 $\Delta\phi = 2\pi \frac{2(R_1 - R_2)}{\lambda}$

Same Range Bin

$$|R_1 - R_2| \leq \delta r$$

- In this instance, the response exhibits a null or dip between the two returns
- Can visually resolve the two Doppler shifts
- In general, composite response falls somewhere between the two extremes

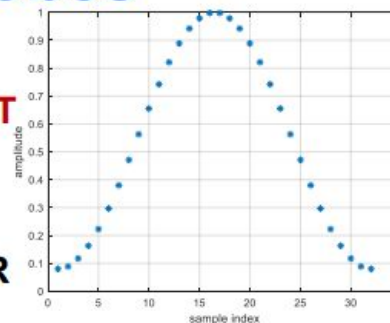


Weighting Reduces Sidelobes

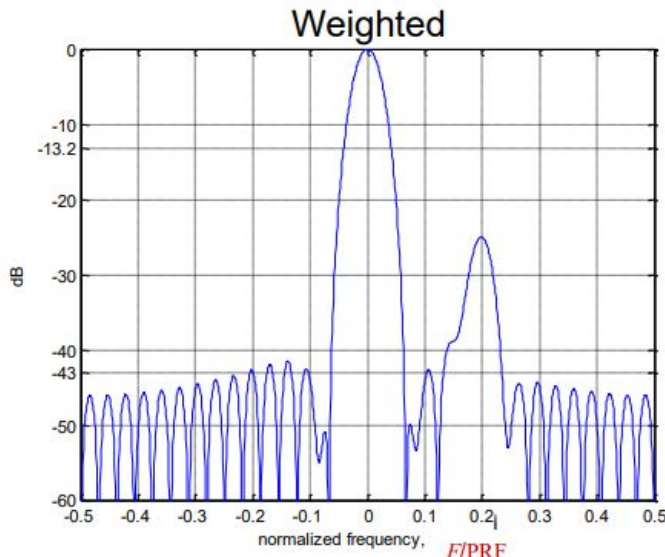
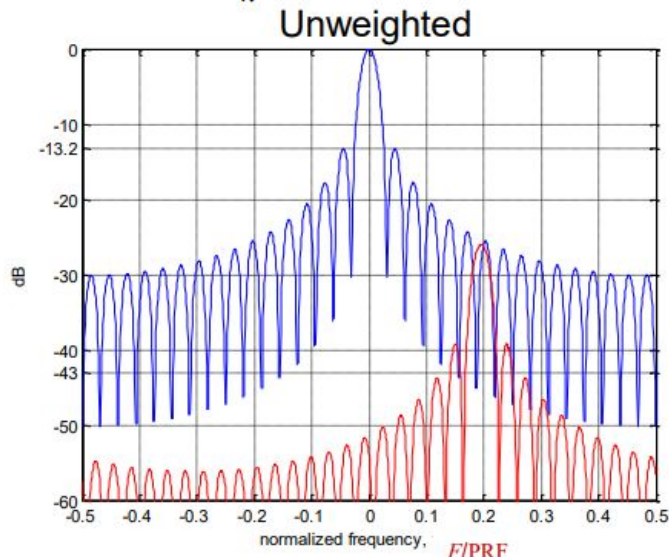
- Sidelobes mask the presence of smaller targets
- Amplitude weighting applied to I/Q samples prior to **DTFT**
 - Reduces sidelobe levels
 - Weighting function defines peak sidelobe level
 - Lowers sidelobes; degrades resolution; reduces SNR

$$X(f) = \sum_{n=-\infty}^{+\infty} w[n]x[n] e^{-j2\pi fn}$$

frequency



Amplitude Weighting

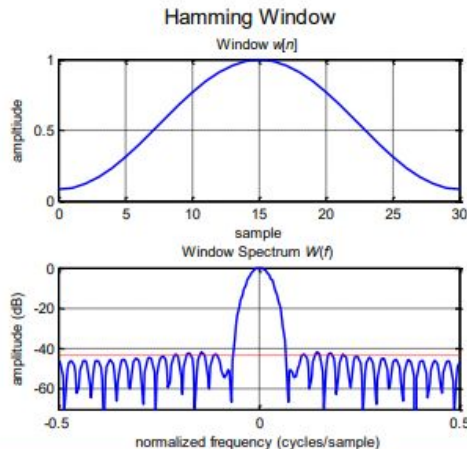


Common Windows

- There are many, many common windows ...

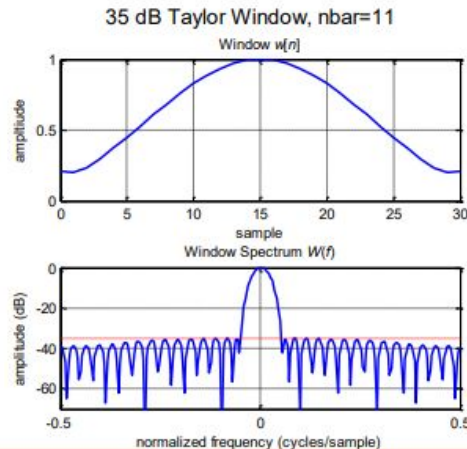
- **Hamming**

- Raised cosine family
- Very simple to compute
 - MATLAB `hamming()`
- Easy to apply in time or frequency
- Not adjustable



- **Taylor**

- Common in SAR and antennas
- Relatively complex calculation
 - MATLAB `taylorwin()`
- Adjustable sidelobe level
- Adjustable sidelobe plateau width



- Others include Hann, Bartlett, Kaiser, Dolph-Chebyshev, Parzen, ...

Taylor Weighting Function

	Peak Sidelobe Level (dB)								
	-20	-25	-30	-35	-40	-45	-50	-55	-60
nbar	SNR Loss (dB)								
2	-0.21	-0.38	-0.51						
3	-0.21	-0.45	-0.67	-0.85					
4	-0.18	-0.43	-0.69	-0.91	-1.11	-1.27			
5	-0.16	-0.41	-0.68	-0.93	-1.14	-1.33	-1.49		
6	-0.15	-0.39	-0.66	-0.92	-1.15	-1.35	-1.53	-1.68	
7	-0.15	-0.37	-0.65	-0.91	-1.15	-1.36	-1.54	-1.71	-1.85
8	-0.16	-0.36	-0.63	-0.90	-1.14	-1.36	-1.55	-1.72	-1.87
9	-0.16	-0.36	-0.63	-0.90	-1.14	-1.36	-1.55	-1.72	-1.87

	Peak Sidelobe Level (dB)								
	-20	-25	-30	-35	-40	-45	-50	-55	-60
nbar	4 dB Resolution Normalized by $c/2\beta$								
2	1.15	1.19	1.21						
3	1.14	1.22	1.28	1.33					
4	1.12	1.22	1.29	1.36	1.42	1.46			
5	1.11	1.20	1.29	1.36	1.43	1.49	1.54		
6	1.10	1.19	1.28	1.36	1.43	1.50	1.56	1.61	
7	1.09	1.19	1.28	1.36	1.43	1.50	1.56	1.62	1.67
8	1.08	1.18	1.27	1.35	1.43	1.50	1.57	1.63	1.68

The Discrete Fourier Transform (DFT) and the Fast Fourier Transform (FFT)

Sampling the DTFT in Frequency: the DFT

- We can compute the spectrum only on a discrete set of frequencies
 - How to sample the DTFT in frequency?
- Consider the discrete sequence formed by taking K samples of one period of the DTFT of a finite-length signal, evenly spaced in frequency:

$$X[k] = X(\omega) \Big|_{\omega = \frac{2\pi k}{K}}, \quad k \in [0, K-1]$$

- Applying this frequency sampling to the DTFT definition gives us the **Discrete Fourier Transform (DFT)**:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/K} \quad x[n] = \frac{1}{K} \sum_{k=0}^{K-1} X[k] e^{+j2\pi nk/K}$$

DFT Indices and Digital Frequency

- DFT index k corresponds to a normalized radian frequency $\omega_k = 2\pi k/K$ radians/sample, or a normalized cyclical frequency $f_k = k/K$ cycles/sample
- Using $\omega = \Omega T = \Omega/F_s$ gives

$$k \rightarrow 2\pi \left(\frac{F_s}{K} \right) k \text{ rad/sec, or } k \rightarrow \left(\frac{F_s}{K} \right) k \text{ Hz}$$

- The total DFT covers F_s Hz, divided into K bins
- So each bin is F_s/K hertz wide

DFT Size and Data Size

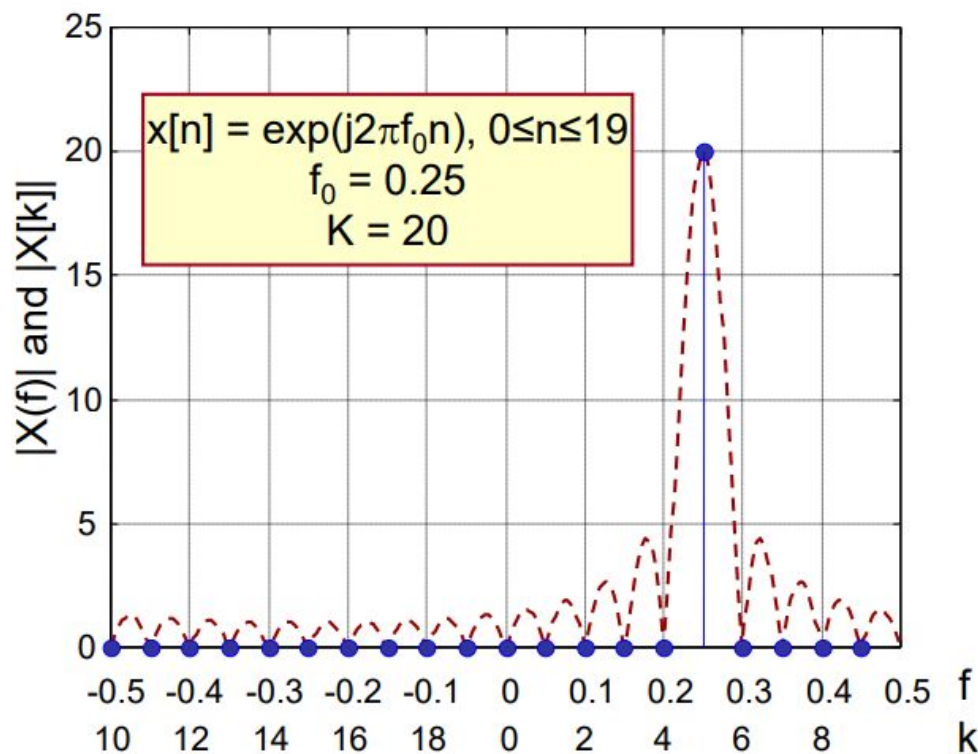
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/K}, \quad k = 0, \dots, K-1$$

- K = DFT size = number of spectrum samples
- K does **not** have to be the same as the number of data samples, N
- Choose K based on the desired sample spacing in frequency and/or requirements for filtering
- Usually $K \geq N$
 - “Zero pad” the data to create a K -point input vector for your FFT software function
- Let's look at the effect of K on the DFT ...

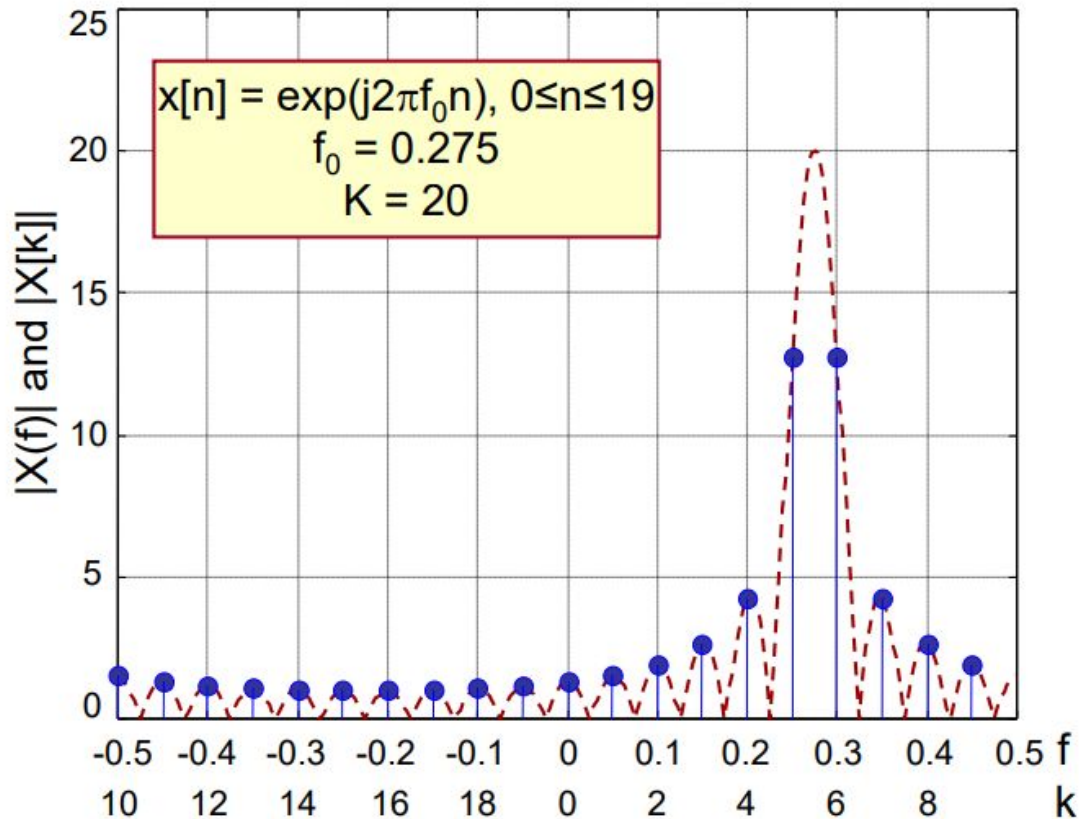
Zero-padding
produces more
closely spaced
frequency
samples
generated by DFT

The DFT Samples the DTFT in Frequency

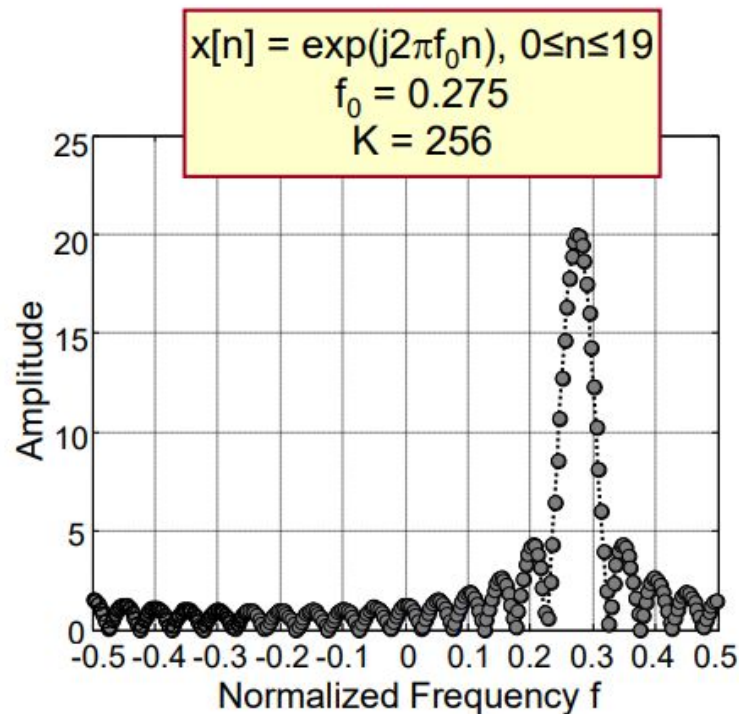
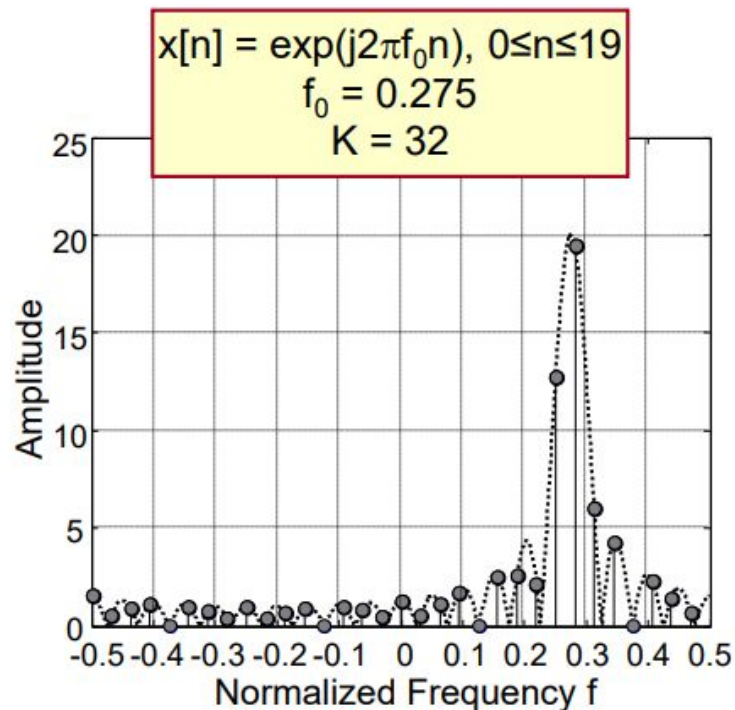
- The data sequence $x[n]$ determines the DTFT
- The DFT samples that DTFT in frequency
- How that works out depends on the number of frequency samples (DFT size K) and how they align with the underlying DTFT
- Here's a favorable example:



A Less Favorable Result



Increasing the DFT Size Tracks the Underlying DTFT More Closely



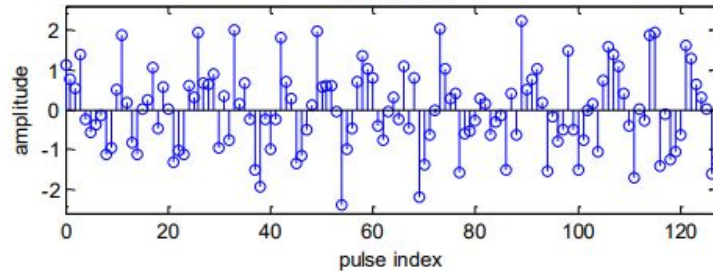
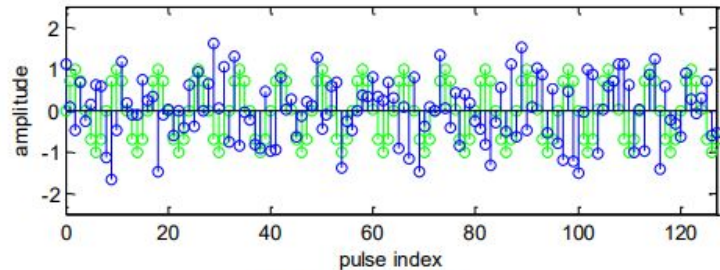
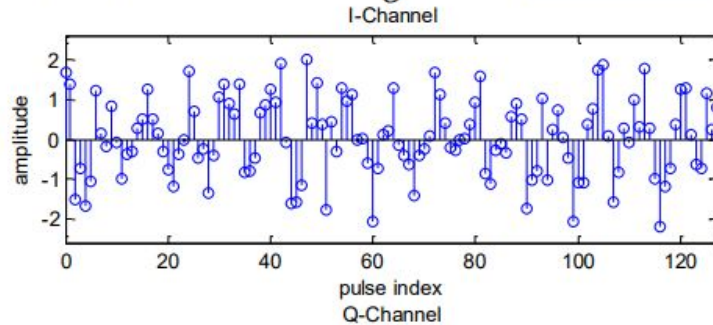
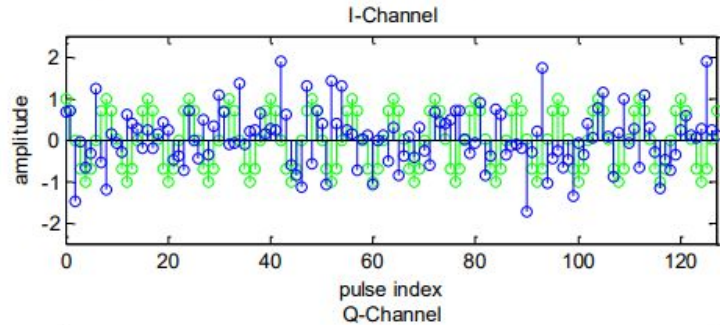
Signal-to-Noise Ratio

Prior to applying the DFT, the SNR may be less than zero

$$x(n) = \exp(j2\pi(0.125PRF)(nT)) \quad n = 0, \dots, (N-1)$$

$$SNR = \frac{A^2}{\sigma^2}$$

SNR = 0 dB



Blue – noise

Green – Doppler shifted signal

Blue – noise + signal

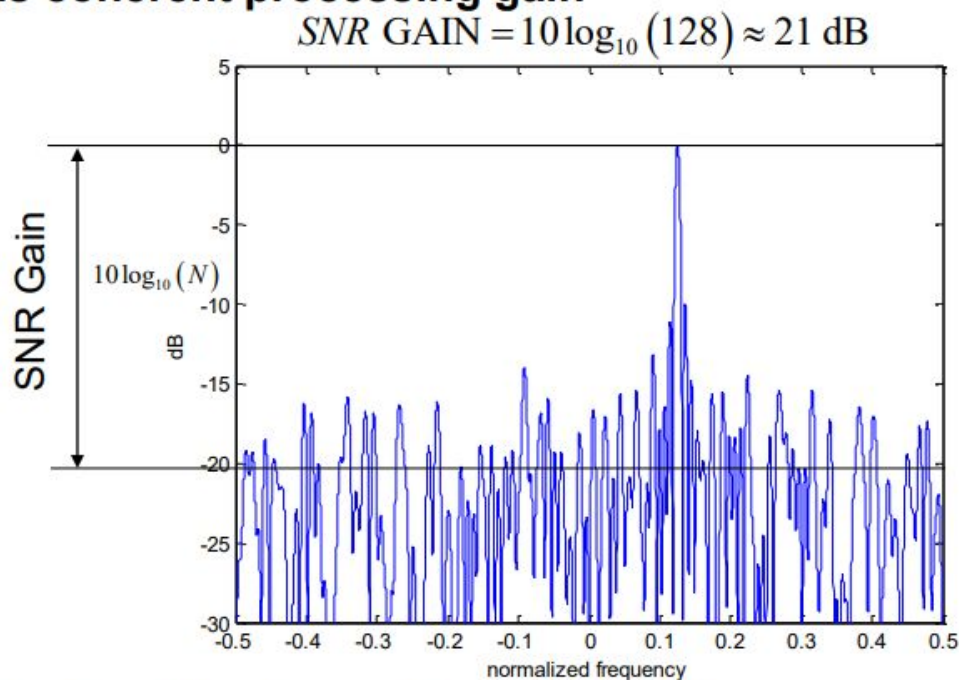
SNR Gain Associated with Doppler Processing

- SNR Gain due to Doppler processing
- Often referred to as coherent processing gain

$$SNR = \frac{A^2}{\sigma^2}$$

$$SNR = \frac{N^2 A^2}{N \sigma^2} = \frac{N A^2}{\sigma^2}$$

$$SNR \text{ GAIN} = 10 \log_{10}(N)$$



- Example of radar modes benefiting from coherent integration
 - SAR: 100s to 1000s of pulses (20 to 30 dB of SNR gain or more)
 - GMTI: 10s to 100s of pulses (10 to 20 dB of SNR gain or more)

The Fast Fourier Transform (FFT)

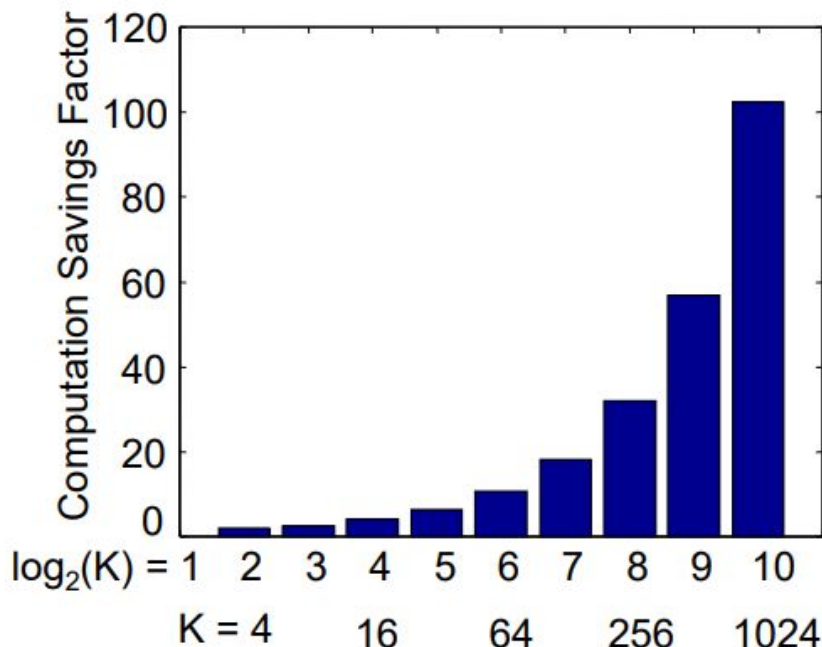
- The FFT **is not** another version of the F.T.
- The FFT **is** an algorithm for efficiently computing the DFT
- The N -point DFT of an N -point sequence $x[n]$ is given by

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/N}, \quad k = 0, \dots, N-1$$

- N multiplies, $N-1$ additions for each of N output samples $\Rightarrow \sim N^2$ complex multiply-adds

Computational Efficiency of the FFT

- For a size N **DFT**, $\sim N^2$ complex multiplications and additions are needed
- For a size N **FFT** (N a power of 2), $\sim N \log_2 N$ multiplications and additions are needed
- Note that the FFT is more of a “win” as the problem size increases



Relation of Fourier Transforms

time	frequency	
Signal Variable	Transform Variable	Transform
continuous	continuous	Fourier Transform (FT)
discrete	continuous	Discrete-Time Fourier Transform (DTFT)
discrete	discrete	Discrete Fourier Transform (DFT)
discrete	discrete	Fast Fourier Transform (FFT)



Sampled data $\Rightarrow$
replicated spectrum

Sampled spectrum $\Rightarrow$
finite data sequence

Fast algorithm
for DFT